

ME371 INTERNAL COMBUSTION ENGINES	
CO1	Study the working of SI and CI Engines and its components.
CO2	Prepare and analyze valve timing diagrams and heat balance sheet.
CO3	Understand and estimate the effects of variation in properties of air at various states.
CO4	Carry out tests on I. C. Engines and calculate its efficiencies and performance parameters at various running conditions.
CO5	Learn about the advanced technologies and research areas in engines.

List of Experiment (ME371 Internal Combustion Engines)		
Sr. No.	Title	Course Outcomes
1	Comparison of S.I and C.I Engines	CO1
2	Combustion Chambers	CO3
3	Types of Carburetor	CO1
4	Ignition System	CO5
5	Fuel Injection System	CO5
6	Single Cylinder Four Stroke Diesel Engine	CO2, CO4
7	Four Cylinder Four Stroke Diesel Engine	CO2, CO4
8	Single Cylinder Four Stroke Petrol Engine	CO2, CO4
9	Single Cylinder Two Stroke Petrol Engine	CO2, CO4
10	Valve Timing Diagram	CO2

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering.

Internal Combustion Engines (ME-371)

INDEX

Sr. No.	Performance Date	Title	No. of Pages	Marks	Assessment Date	Sign of Faculty
1		Comparison of S.I and C.I Engines				
2		Combustion Chambers				
3		Types of Carburetor				
4		Ignition System				
5		Fuel Injection System				
6		Single Cylinder Four Stroke Diesel Engine				
7		Four Cylinder Four Stroke Diesel Engine				
8		Single Cylinder Four Stroke Petrol Engine				
9		Single Cylinder Two Stroke Petrol Engine				
10		Valve Timing Diagram				

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Internal Combustion Engines (ME-371)

Date:

PRACTICAL NO 1

COMPARISON OF S.I AND C.I ENGINES

AIM: To study the comparison between the Spark Ignition engine and Compression Ignition engine

OBJECTIVES:

- (1) To Study the difference between Two Stroke and Four Stroke Engine
- (2) To Study the difference between the S.I and C.I Engine.
- (3) Demonstration on cut section model.

QUESTIONS:

- (1) List the difference between the S.I and C.I engine.
- (2) List the difference between Two Stroke and Four Stroke Engine. Draw a neat sketch showing all components.
- (3) Explain different parts of I C Engine.

Marks obtained:	Signature of faculty:	Date:
------------------------	------------------------------	--------------

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Internal Combustion Engines (ME-371)

Date:

PRACTICAL NO 2
COMBUSTION CHAMBERS

AIM: To Study of Combustion Chambers.

(A) To Study of Combustion Chambers for Spark Ignition Engine

OBJECTIVES:

- (1) To understand combustion process in S.I engines
- (2) To understand different type of combustion chambers

QUESTIONS:

- (1) What are the basic requirements of a good combustion chamber?
- (2) Explain combustion chamber design principal
- (3) Classification of combustion chamber.
- (4) Explain different types of combustion chamber.
- (5) Draw a neat sketch of combustion chamber of 4-stroke and 2-stroke engine.

(B) To Study of Combustion Chamber for Compression Ignition Engine.

OBJECTIVES:

- (1) To understand the combustion processes in C.I. engine.
- (2) To understand the different types of combustion chamber in C.I. Engine.

QUESTIONS:

- (1) Classification of Combustion chamber in case of C.I. Engine.
- (2) Explain different types of combustion chamber in C.I. Engine.

Marks obtained:	Signature of faculty:	Date:
------------------------	------------------------------	--------------

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Internal Combustion Engines (ME-371)

Date:

PRACTICAL NO 3
TYPES OF CARBURETOR

AIM: Study of Different Types of Carburetor in S.I. Engine.

OBJECTIVE:

Understand a carburetion process and learn about different type of carburetors.

QUESTIONS:

- (1) Derive the equation for simple carburetor and put your comments on it.
- (2) Write the shot note on the complete carburetor.
- (3) Write a shot note with neat sketch on the following carburetor.
 - (a) Solex Carburetor
 - (b) S.U. Carburetor
 - (c) Carter Carburetor
- (4) Write a note in brief for Petrol Injection system.

Marks obtained:	Signature of faculty:	Date:
------------------------	------------------------------	--------------

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Internal Combustion Engines (ME-371)

Date:

PRACTICAL NO 4
IGNITION SYSTEM

AIM: Study of Ignition Systems in I.C. Engine.

OBJECTIVE:

Understand ignition systems in I.C engine.

QUESTIONS:

- (1) Explain purpose of Ignition systems, spark timing & spark advance.
- (2) What are the basic requirements of good Ignition system?
- (3) Classification of Ignition system.
- (4) Write a shot note on Buttery Ignition system
- (5) Explain magneto Ignition system.
- (6) Explain electronics Ignition system.
- (7) Construction of spark plug
- (8) What is ignition timing and explain factors affecting Ignition timing.
- (9) Explain (a) Vacuum advanced (b) Centrifugal advanced

Marks obtained:	Signature of faculty:	Date:
------------------------	------------------------------	--------------

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Internal Combustion Engines (ME-371)

Date:

PRACTICAL NO 5
FUEL INJECTION SYSTEM

AIM: Study of Fuel Injection Systems in C. I. Engine.

OBJECTIVE:

Understand Fuel Injection systems & working and construction of Injector and Fuel Pump.

QUESTIONS:

- (1) Requirements of good diesel Injection system.
- (2) Types of Injection system
- (3) Explain types of Nozzles
- (4) Write a short note of Fuel Pump (working and construction with neat sketch)

Marks obtained:	Signature of faculty:	Date:
------------------------	------------------------------	--------------

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Internal Combustion Engine (ME-371)

Date:

PRACTICAL NO 6
SINGLE CYLINDER FOUR STROKE DIESEL ENGINE

AIM: Performance test on four-stroke single-cylinder diesel engine with eddy current dynamometer.

OBJECTIVE:

To perform test on four-stroke single cylinder diesel engine with eddy current dynamometer to understand its operational characteristics.

SPECIFICATIONS:

Type:	Four Stroke Single Cylinder Diesel Engine
Make:	Kirloskar model JAV1
Bore:	85 mm
Stroke:	80 mm
RPM:	1500
BHP:	5 HP, 3.7 KW
Fuel:	High Speed Diesel oil
Sp. Gr.:	0.83
C.V:	10,833 Kcal/kg

PRECAUTIONS:

- It is strongly recommended that the operator is familiar with the engine before it is started. Before the engine is started, check the lubricating oil level I the crank case and add oil if required. All recommended periodic maintenance and service in the manual must be carried out for the engine.
- At any time either the radiator or the external cooling water line is to be used. Both cannot be used at the same time. When external cooling water is used, radiator valves should be closed to avoid leakage or even rupture due to high-pressure water supply.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

- Do not operate the exhaust gas calorimeter without water as the brazing used for copper tubes in the heat exchanger will melt at full load engine exhaust temperature (over 400-500 deg C)

PROCEDURE:

- Check fuel level.
- Check lubrication oil level.
- Open the three way cock so that fuel flows to the engine directly from the tank.
- Open the cooling water valves and ensure that water flows through the engine.
- Open the water line to the hydraulic dynamometer.
- Keep the loading in the hydraulic dynamometer at minimum.
- Start the engine. If the engine is cold, press the heater button until the glow indicator glows and then start the engine.
- Operate the throttle valve so that the engine picks up the speed to the required level, say 1500 rpm.
- Load engine with the hydraulic dynamometer loading is achieved by turning the handle in the direction marked. If sufficient load is not absorbed by the dynamometer at the required speed, the outlet valve in the dynamometer can be closed to increase the pressure (as indicated by the pressure gauge) and hence the load.
- When engine is loaded the speed will decrease. Hence open the throttle to increase the speed.
- When steady condition is reached, the cooling water temperature is maintained at the required level by adjusting the flow rate. Measure the flow rate.
- Adjust the cooling water flow rate in the exhaust gas calorimeter to achieve steady state condition.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

OBSERVATION TABLE:

Sr. No.	Particulars	1	2	3	4	5
1	Engine Speed (N rpm)					
2	Hydraulic Dynamometer Load (W kg)					
3	Engine Output (KW)					
4	Time for 10 cc of fuel consumption (Sec)					
5	Engine input (KW)					
6	Thermal Efficiency %					

Air Tank Measurement:

Orifice Water Manometer Reading h_1 = _____ cm of water

Orifice Water Manometer Reading h_2 = _____ cm of water

Difference in water level Δh = _____ cm of water

Exhaust gas calorimeter readings:

Engine cooling water inlet temperature T_1 = _____

Engine cooling water outlet temperature T_2 = _____

Calorimeter cooling water inlet temperature T_3 = _____

Calorimeter cooling water outlet temperature T_4 = _____

Exhaust gas inlet temperature T_5 = _____

Exhaust gas outlet temperature T_6 = _____

Room Temperature T_7 = _____

Engine cooling water flow rate W_e = _____

Calorimeter cooling water flow meter W_c = _____

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Length of torque arm $L = 0.146 \text{ m}$
 Load $W = \underline{\hspace{2cm}}$

Torque $T = 9.81 \times W \times L = \underline{\hspace{2cm}}$

Engine speed $N = \underline{\hspace{2cm}}$

Engine output $= \frac{2 \times 3.14 \times N \times 9.81 \times W \times L}{1000 \times 60} = \underline{\hspace{2cm}} \text{ KW}$

Time for 10cc of fuel consumption $t = \underline{\hspace{2cm}}$

Fuel consumption per minute $Q = \frac{10}{t} \times 60 = \underline{\hspace{2cm}} \text{ cc/min}$

Total fuel consumption = $\frac{Q \times \text{Sp.Gr of diesel}}{1000} = \underline{\hspace{2cm}} \text{ kg/min}$

Heat input = total fuel consumption $\times$ C.V = $\underline{\hspace{2cm}} \text{ Kcal/min}$

Input power = *heat input* / 14.34 KW = $\underline{\hspace{2cm}}$

Engine efficiency = $\frac{\text{engine output}}{\text{engine input}} \times 100 = \underline{\hspace{2cm}}$

Equivalent air column = $\frac{\rho_w}{\rho_a} \times \Delta h = \underline{\hspace{2cm}}$

Orifice diameter = $d = 0.035 \text{ m}$

Area of orifice = $0.962 \times 10^{-3} \text{ m}^2$

Volume of air = $V_a = C_d \times \text{Area of orifice} \times (2gh)^{0.5} = \underline{\hspace{2cm}}$

Weight of intake air = $1.2 \times V_a = \underline{\hspace{2cm}} \text{ kg/sec}$

Theoretical volume of air = $V_t = \frac{\pi/4 \times D^2 \times L \times N \times 4}{2 \times 60} = \underline{\hspace{2cm}}$

Volumetric efficiency = $\frac{V_a}{V_t} \times 100 = \underline{\hspace{2cm}}$

Air fuel ratio = $\underline{\hspace{2cm}}$

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

GRAPH:

Draw the graph of η Vs Load, Brake power

CONCLUSION:

Marks obtained:	Signature of faculty:	Date:
------------------------	------------------------------	--------------

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Internal Combustion Engine (ME-371)

Date:

PRACTICAL NO 7

FOUR CYLINDER FOUR STROKE DIESEL ENGINE

AIM: Performance test on four-stroke four-cylinder diesel engine with hydraulic dynamometer.

OBJECTIVE:

To perform test on four-stroke four cylinders diesel engine with hydraulic dynamometer to understand its operational characteristics.

SPECIFICATIONS:

Type:	Four Stroke Four Cylinder Diesel Engine
Make:	HM Trekker
Bore:	73 mm
Stroke:	90 mm
Capacity:	1500 cc
RPM:	1500
BHP:	10 HP @ 1500 rpm
Fuel:	High Speed Diesel oil
Sp. Gr.:	0.83
C.V:	10,833 Kcal/kg

PRECAUTIONS:

- It is strongly recommended that the operator is familiar with the engine before it is started. Before the engine is started, check the lubricating oil level in the crankcase and add oil if required. All recommended periodic maintenance and service in the manual must be carried out for the engine.
- At any time either the radiator or the external cooling water line is to be used. Both cannot be used at the same time. When external cooling water is used, radiator valves should be closed to avoid leakage or even rupture due to high-pressure water supply.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

- Do not operate the exhaust gas calorimeter without water as the brazing used for copper tubes in the heat exchanger will melt at full load engine exhaust temperature (over 400-500 deg C)

PROCEDURE:

- Check fuel level.
- Check lubrication oil level.
- Open the three way cock so that fuel flows to the engine directly from the tank.
- Open the cooling water valves and ensure that water flows through the engine.
- Open the water line to the hydraulic dynamometer.
- Keep the loading in the hydraulic dynamometer at minimum.
- Start the engine. If the engine is cold, press the heater button until the glow indicator glows and then start the engine.
- Operate the throttle valve so that the engine picks up the speed to the required level, say 1500 rpm.
- Load engine with the hydraulic dynamometer loading is achieved by turning the handle in the direction marked. If sufficient load is not absorbed by the dynamometer at the required speed, the outlet valve in the dynamometer can be closed to increase the pressure (as indicated by the pressure gauge) and hence the load.
- When engine is loaded the speed will decrease. Hence open the throttle to increase the speed.
- When steady condition is reached, the cooling water temperature is maintained at the required level by adjusting the flow rate. Measure the flow rate.
- Adjust the cooling water flow rate in the exhaust gas calorimeter to achieve steady state condition.

OBSERVATION TABLE:

Sr. No.	Particulars	1	2	3	4	5
1	Engine Speed (N rpm)					
2	Hydraulic Dynamometer Load (W kg)					
3	Engine Output (KW)					
4	Time for 10 cc of fuel consumption (Sec)					
5	Engine input (KW)					
6	Thermal Efficiency %					

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

CALCULATION:

Air Tank Measurement:

Orifice Water Manometer Reading h1= _____ cm of water

Orifice Water Manometer Reading h2= _____ cm of water

Difference in water level Δh = _____ cm of water

Exhaust gas calorimeter readings:

Engine cooling water inlet temperature T1= _____

Engine cooling water outlet temperature T2= _____

Calorimeter cooling water inlet temperature T3= _____

Calorimeter cooling water outlet temperature T4= _____

Exhaust gas inlet temperature T5= _____

Exhaust gas outlet temperature T6= _____

Room Temperature T7= _____

Engine cooling water flow rate We= _____

Calorimeter cooling water flow meter Wc= _____

Hydraulic dynamometer load W= _____

Engine speed N= _____

Engine output
$$= \frac{WN}{2000 \times 1.36} = \text{_____} \text{ KW}$$

Time for 10cc of fuel consumption t = _____

Fuel consumption per minute $Q = \frac{10}{t} \times 60 = \text{_____} \text{ cc/min}$

Total fuel consumption =
$$\frac{Q \times \text{Sp.Gr of diesel}}{1000} = \text{_____} \text{ kg/min}$$

Heat input = total fuel consumption $\times$ C.V = _____ Kcal/min

Input power = heat input / 14.34 KW = _____

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

$$\text{Engine efficiency} = \frac{\text{engine output}}{\text{engine input}} \times 100 = \underline{\hspace{2cm}}$$

$$\text{Equivalent air column} = \frac{\rho_w}{\rho_a} \times \Delta h = \underline{\hspace{2cm}}$$

$$\text{Orifice diameter} = d = 0.035 \text{ m}$$

$$\text{Area of orifice} = 0.962 \times 10^{-3} \text{ m}^2$$

$$\text{Volume of air} = V_a = C_d \times \text{area of orifice} \times (2gh)^{0.5} = \underline{\hspace{2cm}}$$

$$\text{Weight of intake air} = 1.16 \times V_a = \underline{\hspace{2cm}} \text{ kg/sec}$$

$$\text{Theoretical volume of air} = V_t = \frac{\pi/4 \times D^2 \times L \times N \times 4}{2 \times 60} = \underline{\hspace{2cm}}$$

$$\text{Volumetric efficiency} = \frac{V_a}{V_t} \times 100 = \underline{\hspace{2cm}}$$

$$\text{Air fuel ratio} = \underline{\hspace{2cm}}$$

GRAPH:

Draw the graph of η Vs Load, Brake power

CONCLUSION:

Marks obtained:	Signature of faculty:	Date:
------------------------	------------------------------	--------------

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Internal Combustion Engine (ME-371)

Date:

PRACTICAL NO 8
SINGLE CYLINDER FOUR STROKE PETROL ENGINE

AIM: Performance test on four-stroke single-cylinder petrol engine with eddy current dynamometer.

OBJECTIVE:

To perform test on four-stroke single cylinder petrol engine with eddy current dynamometer to understand its operational characteristics.

SPECIFICATIONS:

Make: Greaves

Type: four stroke single cylinder vertical petrol engine

Power: 2.9 KW

Speed: 3000 rpm

PRECAUTION:

- Check water supply and crankcase oil level starting the engine.
- Never stop the engine on load
- Do not put hand in water rheostat while in operation
- Do not close the orifice while engine is in operation
- Add salt in water rheostat incorrect percentage
- After completion of experiment drain all water

PROCEDURE:

- Fill up sufficient petrol in tank without oil
- Check oil level in the crankcase of the engine. It is lesser than H on dipstick level it up to the mark with clean SAE-40oil
- Fill up water in water rheostat to a required height and add salt. (Approximately 50 gm in filled water tank)
- Fill up water in manometer up to half of the height. Put off main switch o rheostat and ignition switch o engine is ON

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

- Start water supply to calorimeter
- Wind the starting rope over the starting drum and give a brisk pull. Initially, it may require operating the choke, if the engine is cold. As the engine starts put the choke off. Put on main switch.
- Slowly rotate the loading knob (on panel) in clockwise direction so that the engine gets loaded, say 2 amps
- Fill the burette with burette filling cock. Close the cock in fuel line coming from fuel tank note down time for 10 ml fuel consumption, manometer difference, ammeter and voltmeter reading and calorimeter water flow
- Note down the temperature from T1 to T4
- Repeat the procedure for difference loads. Maximum load is 8 amps.

OBSERVATION TABLE:

Sr. No	1	2	3	4
Brake Drum Speed N rpm				
Ammeter I amp				
Voltmeter V volt				
Time for 10 ml of fuel t_f sec				
Manometer diff. h_w cm				
Time for 1 lit. Calorimeter water t_w sec				
Temperatures:				
Water inlet to calorimeter T_1 deg C				
Water outlet from calorimeter T_2 deg C				
Exhaust gas before calorimeter T_3 deg C				
Exhaust gas after calorimeter T_4 deg C				

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

CALCULATION:

Brake power

$$BP = \frac{V \times I}{0.7} + 0.5 + \left(\frac{V}{220} \times 0.198 \right) = \text{_____} \text{ KW}$$

Note: practically dynamometer efficiency also varies with load and for each load, different efficiency has to be taken from dynamometer efficiency curve, but for to avoid complications, same efficiency is assumed.

Field coil is also energized from armature, which consumes 0.198 kw at 220 volts. Hence, the load is added to alternator with voltage correction factor.

For the tests conducts on the blower a the factory, it is observed that blower consumes power of 0.5 kw at 3000 rpm

Fuel consumption

$$FC = (10 / tf) \times (3600 \times 0.7 / 1000) = \text{_____} \text{ kg/hr}$$

Density of petrol is 0.7 gms/cc

Specific fuel consumption

$$SFC = FC / BP = \text{_____} \text{ kg/kwhr}$$

Heat supplied by fuel

$$H_f = FC \times 43890 = \text{_____} \text{ KJ/hr}$$

C. V of petrol is 43890 KJ/kg

Heat equivalent to BP

$$H_{BP} = BP \times 3600 = \text{_____} \text{ KJ/hr}$$

Efficiency

$$\text{Brake thermal efficiency} = \frac{\text{heat equivalent to BP}}{\text{heat supplied by fuel}} \times 100 = \text{_____}$$

Air consumption

$$h_a = h_w (1000 / \rho_a)$$

$$= p / 287 \times T = \text{_____}$$

Mass flow of air

$$m_a = 0.62 \times 1.54 \times 10^{-4} \times (2gh)^{0.5} \times 3600 \times \rho_a = \text{_____}$$

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

where, C_d of orifice = 0.62

Area of orifice = $1.54 \times 10^{-4} \text{ m}^2$

Air fuel ratio

AFR = m_a/FC = _____

Heat carried away by exhaust gases

Heat given by exhaust gases in calorimeter = heat gained by water

$$m_{eg} \times C_{peg} \times (T_3 - T_4) = m_w \times C_{pw} \times (T_2 - T_1)$$

m_{eg} = mass flow of exhaust gases

C_{peg} = specific heat of exhaust gases

m_w = calorimeter water flow rate, Kg/hr

C_{pw} = specific heat of water 4.18 KJ/kg K

Heat carried away by exhaust gases, on ambient temperature basis

$$H_{eg} = (m_g \times C_{peg}) \times (T_3 - T_1) = \text{_____}$$

GRAPH:

Plot the graph of BP Vs efficiency, FC and SFC

CONCLUSION:

Marks obtained:	Signature of faculty:	Date:
------------------------	------------------------------	--------------

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Internal Combustion Engine (ME-371)

Date:

PRACTICAL NO 9
SINGLE CYLINDER TWO STROKE PETROL ENGINE

AIM: Performance test on two-stroke single-cylinder petrol engine with rope brake dynamometer

OBJECTIVE:

To perform test on two-stroke single cylinder petrol engine with rope brake dynamometer to understand its operational characteristics.

SPECIFICATIONS:

Make: Bajaj
Type: two stroke single cylinder horizontal petrol engine
Power: 4.5 HP
Speed: 6000 rpm

PRECAUTIONS:

- Check water supply and crankcase oil level before starting the engine.
- 2% 2T oil must be added to petrol
- Never stop the engine on load
- From first 30 hrs of engine operation do not increase engine speed above 3000 rpm

PROCEDURE:

- Fill up sufficient petrol in tank with 2% self mixing 2T oil
- Check oil level in the gearbox of the engine. If necessary add up SAE-40 oil. Oil level should be up to the oil-filling hole.
- Confirm that the engine is in neutral gear and ignition switch in ON
- Start water supply to calorimeter and brake drum
- Pull the choke knob and give a sharp kick, engine will start. As the engine starts, release the choke knob, pull the clutch and put the engine in 4th gear
- Slowly increase the accelerator and set the engine speed to say 2500 rpm
- Put load on the engine.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

- Fill the burette with burette filling cock. Note down time for 10ml fuel consumption, manometer difference, spring balance difference, calorimeter water flow rate and brake drum speed.
- Note down and temperatures from T₁ to T₄
- Increase the load by twisting the loading screw. Keep the speed constant with the help of accelerator
- Repeat the procedure for the different loads
- Now keep accelerator position constant and go on increasing the load and note down the speed and fuel consumption at each load

OBSERVATION TABLE:

Sr. No	1	2	3	4
Brake Drum Speed N rpm				
Ammeter I amp				
Voltmeter V volt				
Time for 10 ml of fuel t _f sec				
Manometer diff. h _w cm				
Time for 1 lit. Calorimeter water t _w sec				
Temperatures:				
Water inlet to calorimeter T ₁ deg C				
Water outlet from calorimeter T ₂ deg C				
Exhaust gas before calorimeter T ₃ deg C				
Exhaust gas after calorimeter T ₄ deg C				

CALCULATION:

Brake power

T= load × 9.81 × effective brake radius = _____

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

$$BP = \frac{(2 \times 3.14 \times N \times T)}{60000 \times 0.95} + \text{blower power KW} = \underline{\hspace{2cm}}$$

Note:

The engine is not directly coupled to rope brake dynamometer but through gearbox, power loss is there, which is taken as 5% i.e., transmission efficiency is assumed to be 95%.

Fuel consumption

$$FC = (10 / tf) \times (3600 \times 0.7 / 1000) = \underline{\hspace{2cm}} \text{ kg/hr}$$

Density of petrol is 0.7 gms/cc

Specific fuel consumption

$$SFC = FC / BP = \underline{\hspace{2cm}} \text{ kg/kwhr}$$

Heat supplied by fuel

$$H_f = FC \times 43890 = \underline{\hspace{2cm}} \text{ KJ/hr}$$

C. V of petrol is 43890 KJ/kg

Heat equivalent to BP

$$H_{BP} = BP \times 3600 = \underline{\hspace{2cm}} \text{ KJ/hr}$$

Efficiency

$$\text{Brake thermal efficiency} = \frac{\text{heat equivalent to BP}}{\text{heat supplied by fuel}} \times 100 = \underline{\hspace{2cm}}$$

Air consumption

$$h_a = (hw/100) \times (1000/\rho_a) \\ = p/287 \times T = \underline{\hspace{2cm}}$$

Mass flow of air

$$m_a = 0.62 \times 1.54 \times 10^{-4} \times (2gh)^{0.5} \times 3600 \times \rho_a = \underline{\hspace{2cm}}$$

where C_d of orifice = 0.62

$$\text{area of orifice} = 1.54 \times 10^{-4} \text{ m}^2$$

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Air fuel ratio

$$AFR = \frac{m_a}{F_C} = \underline{\hspace{2cm}}$$

Heat carried away by exhaust gases

Heat given by exhaust gases in calorimeter = heat gained by water

$$m_{eg} \times C_{peg} \times (T_3 - T_4) = m_w \times C_{pw} \times (T_2 - T_1)$$

m_{eg} = mass flow of exhaust gases

C_{peg} = specific heat of exhaust gases

m_w = calorimeter water flow rate, Kg/hr

C_{pw} = specific heat of water 4.18 KJ/kg K

Heat carried away by exhaust gases, on ambient temperature basis

$$H_{eg} = (m_g \times C_{peg}) \times (T_3 - T_1) = \underline{\hspace{2cm}}$$

GRAPH:

Plot the graph of BP Vs efficiency, FC and SFC

CONCLUSION:

Marks obtained:	Signature of faculty:	Date:
------------------------	------------------------------	--------------

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Internal Combustion Engine (ME-371)

Date:

PRACTICAL NO 10
VALVE TIMING DIAGRAM

AIM: To draw valve timing diagram on four Stroke diesel and four stroke petrol engine

OBJECTIVE:

To understand actual valve timing diagram.

SPECIFICATIONS:

Petrol Engine:

Make:

Type:

Power:

Speed:

Diesel Engine:

Make:

Type:

Power:

Speed:

QUESTIONS:

- (1) Why actual valve timing diagram is different than theoretical valve timing diagram?
- (2) Explain valve timing diagram for low and high-speed four stroke engines with figure.
- (3) Explain valve timing diagram for two stroke engines with figure.

Marks obtained:	Signature of faculty:	Date:
------------------------	------------------------------	--------------